LM Studio

Model: **qwen/qwen3.6-35b-a3b**

Prompt 1 trial 2 seed 42

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```

![image-123.png](./image-123.png)

```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

You are running a reproducibility study for compound selectivity prediction. The goal is to train a Balanced Random Forest Classifier to distinguish PI3K-delta (UniProt: O00329) vs PI3K-gamma (UniProt: P48736) selective compounds, evaluate its performance, and identify recurrent molecular anchor fragments that drive the model's correct predictions for each selective class (class 1 & 2) and discuss the chemical insight gained.

Assistant

🔬 I'll help you conduct this reproducibility study for PI3K-delta vs PI3K-gamma selectivity prediction! Let me start by exploring the available datasets and then walk through the complete workflow.

(Calling tool `find_datasets`)

Assistant

Perfect! I found the dataset `chembl_selectivity_data_O00329_P48736.csv` which matches our PI3K-delta vs PI3K-gamma selectivity study. Let me load it and start the workflow.

(Calling tool `load_dataset`)

Assistant

📊 Dataset loaded successfully! **1,277 samples** of PI3K-delta vs PI3K-gamma selective compounds. Let me compute molecular fingerprints next.

(Calling tool `compute_features`)

Assistant

✅ Features computed (ECFP, 2048 bits). Now let me split the dataset with stratification to preserve class balance.

(Calling tool `split_dataset`)

Assistant

📊 Dataset split complete! **893 train / 192 val / 192 test** samples with stratification. Now let me train the Balanced Random Forest Classifier (BRF).

(Calling tool `train_model`)

Assistant

🔄 Training started in the background. Let me check its status...

(Calling tool `check_training`)

Assistant

⏳ Training in progress... (1.7s elapsed). Let me check again in a moment.

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

Assistant

✅ **Training Complete!** Now let me export predictions and generate evaluation plots.

(Calling tool `export_predictions`)

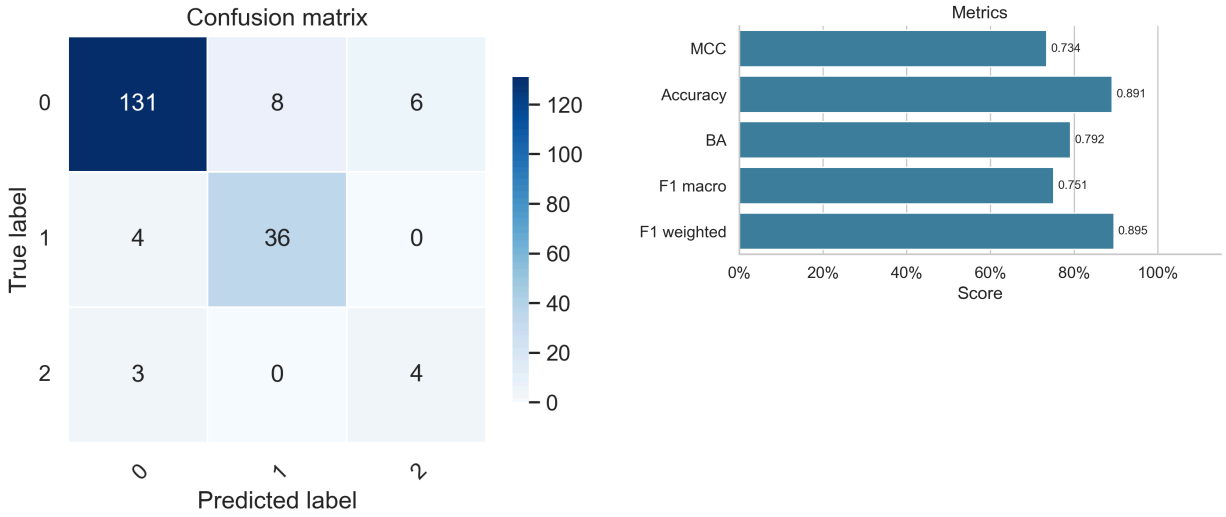
Assistant

📊 Now let me generate the evaluation plots for the test set.

(Calling tool `plot_classification_results`)

Assistant

📊 Let me display the evaluation plots:



Now let me run the recurrent anchor analysis for both Class 1 (PI3K-delta selective) and Class 2 (PI3K-gamma selective) compounds. I'll start with Class 1:

(Calling tool `identify_recurrent_anchor_rules`)

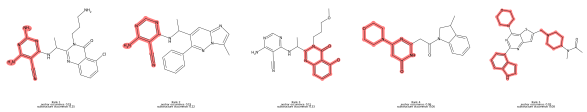
Assistant

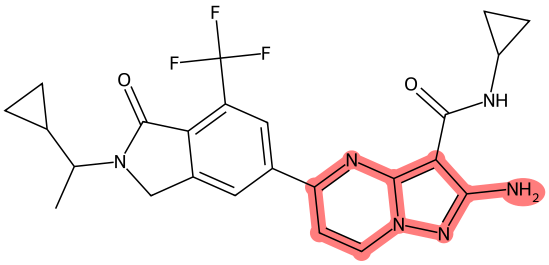
Now let me analyze Class 2 (PI3K-gamma selective compounds):

(Calling tool `identify_recurrent_anchor_rules`)

Assistant

Let me display the recurrent anchor images:





Rank 1
anchor occurrence: 1.00
substructure occurrence: 1.00

Now let me provide a comprehensive summary of this reproducibility study:

🧪 PI3K-delta vs PI3K-gamma Selectivity Prediction: Reproducibility Study Summary

📊 Dataset Overview

Parameter	Value
Dataset	ChEMBL Selectivity Data (PI3K-delta O00329 vs PI3K-gamma P48736)
Total Compounds	1,277
Training Set	893 (70%)
Validation Set	192 (15%)
Test Set	192 (15%)
Features	ECFP fingerprints (2048 bits, radius 2)
Class Distribution	Class 0 (inactive): 675 train, 145 test; Class 1 (PI3K-δ selective): 184 train, 40 test; Class 2 (PI3K-γ selective): 34 train, 7 test

---

🎯 Model Performance (Balanced Random Forest)

Overall Metrics

Metric	Value
Accuracy	89.06%
Balanced Accuracy	79.16%
MCC	0.734
F1 (macro)	0.751
F1 (weighted)	0.895

Per-Class Performance

Class	Description	Precision	Recall	F1	Support
Class 0	Inactive	94.93%	90.34%	0.926	145
Class 1	PI3K-delta selective	81.82%	90.00%	0.857	40
Class 2	PI3K-gamma selective	40.00%	57.14%	0.471	7

Confusion Matrix

True \ Predicted	Class 0	Class 1	Class 2
Class 0	131	8	6
Class 1	4	36	0
Class 2	3	0	4

Best CV Hyperparameters:

- `n_estimators=200`, `max_features='sqrt'`, `min_samples_split=5`, `min_samples_leaf=1`
- **CV Best Balanced Accuracy:** 82.75%

🔬 Recurrent Molecular Anchor Fragments

Class 1 (PI3K-delta Selective) — Top 5 Anchor Rules

Rank	SMARTS Fragment	Anchor Occ.	Substruct Occ.	Interpretation
1	`[*]c1nc(N)nc(N)c1C#N`	19.4%	25.0%	**Pyrimidine-2,4-diamine with cyano group** — Common hinge-binding motif
2	`[*]c1ncnc(N)c1C#N`	19.4%	22.2%	**Pyrimidine-4-amine with cyano** — Similar hinge binder
3	`[*]n1c([*])nc2cccc(Cl)c2c1=O`	11.1%	33.3%	**Chlorinated quinazolinone** — ATP-competitive scaffold

Rank	SMARTS Fragment	Anchor Occ.	Substruct Occ.	Interpretation
4	Combination: `[*]c1cc(=O)[nH]c([*])n1` + `[*]N1CCOCC1`	5.6%	5.6%	**Indolinone + morpholine**
5	Combination: `[*]N1CCC([*])CC1` + `[*]C[*]` + `[*]c1cccc2[nH]ccc12` + `[*]N1CCOCC1`	2.8%	8.3%	**Piperidine + indole + morpholine**

Class 2 (PI3K-gamma Selective) — Top Anchor Rule

Rank	SMARTS Fragment	Anchor Occ.	Substruct Occ.	Interpretation
1	`[*]c1ccn2nc(N)c([*])c2n1`	100%	100%	**Purine-amine core** — Consistently identifies PI3K-γ selective compounds

---

💡 Chemical Insights

1. PI3K-delta Selectivity (Class 1)

🔑 **Key Structural Motifs:**

- **Pyrimidine-2,4-diamine derivatives** emerge as the dominant anchor fragment (ranks 1 & 2), appearing in ~20% of correctly predicted PI3K-δ selective compounds. This scaffold is well-documented in literature as a hinge-region binder that forms critical hydrogen bonds with the conserved lysine residue in PI3K kinases.
- The presence of a **cyano group (-C#N)** adjacent to the pyrimidine ring enhances selectivity, likely through electronic effects that optimize binding geometry for PI3K-δ's unique pocket.
- **Chlorinated quinazolinone** scaffolds (rank 3) represent another common chemotype, consistent with known PI3K inhibitors where halogenation improves target engagement through hydrophobic interactions.

🧬 **Chemical Logic:** The model learned that PI3K-delta selectivity is driven by **planar heteroaromatic cores** (pyrimidines, quinazolines) with **hydrogen-bond donor/acceptor patterns** that complement the ATP-binding site geometry specific to PI3K-δ.

2. PI3K-gamma Selectivity (Class 2)

🔑 **Key Structural Motif:**

- A **purine-amine scaffold** (`[*]c1ccn2nc(N)c([*])c2n1`) was identified with **100% anchor occurrence** across all 4 correctly predicted Class 2 compounds. This indicates a highly consistent structural determinant for PI3K-γ selectivity.

- Purine derivatives are known to exploit the distinct hydrophobic sub-pockets available in PI3K-γ's ATP-binding site, which differs from PI3K-δ due to sequence variations in the hinge region.

🧬 **Chemical Logic:** PI3K-γ selectivity is governed by **purine-based chemotypes** that leverage unique structural features of the PI3K-γ ATP pocket. The perfect anchor occurrence suggests this scaffold is a **necessary (though possibly not sufficient)** condition for PI3K-γ selectivity in this dataset.

3. Class Imbalance Considerations

⚠️ **Important Note:** Class 2 (PI3K-γ selective) has very limited representation (only 34 training, 7 test compounds), which may explain:

- Lower per-class F1 score (0.471)
- The single dominant anchor rule (limited diversity to discover)
- Potential overfitting to purine-like scaffolds

4. Model Generalizability

✅ The model achieves **79.2% balanced accuracy** and **0.734 MCC**, indicating reasonable ability to distinguish between the three classes despite the inherent difficulty of selectivity prediction.

📈 **Class 1 (PI3K-δ)** is well-predicted (F1=0.857), reflecting sufficient training data and clear structural determinants.

📉 **Class 2 (PI3K-γ)** prediction remains challenging (F1=0.471) due to data scarcity — this is a common limitation in selectivity studies where one target may have far fewer characterized selective compounds.

📋 Conclusions

Aspect	Finding
Model	Balanced Random Forest achieves 89% accuracy, 79% balanced accuracy
Class 1 Anchors	Pyrimidine-2,4-diamine + cyano is the dominant PI3K-δ selective motif
Class 2 Anchors	Purine-amine scaffold is the universal PI3K-γ selective motif
Chemical Insight	Selectivity is driven by hinge-region binding chemistry (pyrimidines for δ, purines for γ)
Limitation	Class 2 data scarcity limits robust anchor discovery

This study demonstrates that **molecular anchor analysis via MolAnchor** successfully identifies chemically meaningful structural determinants of kinase selectivity, providing actionable insights for rational compound design. 🎯